

Scalability & Future Plan

Date	1 October 2026
Team ID	SWTID-2026-2533
Project Name	EduGenie Google Gemini Powered Learning Assistant
Maximum Marks	1 Mark

Scalability & Future Plan:

This document explains how EduGenie can grow to serve more learners and handle more features, and records the team's roadmap beyond the current release.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The AI features depend on one Gemini API key with limited quota	Many learners at once can hit rate limits and see errors	High
2	No user accounts or stored history	Learners lose past answers, quiz results and learning paths when the page closes	High
3	The local explanation model is large (about 3 GB) and slow to load	Hard to run on small machines or serverless hosting, so explanations fall back to Gemini there	Medium
4	Answers come only from AI generation and are not linked to a specific syllabus	Learning paths may not match a learner's school or college curriculum	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Runs on Vercel with one shared API key; every request calls the AI.	Cache answers for common questions, add rate limiting per visitor, and use a paid or multiple API quota.
2	Data Storage	Stateless: nothing is saved.	Add a database (SQLite, later PostgreSQL) to store history, quiz scores and learning paths per user.
3	Performance	AI responses are generated on every request; local model loads on first use.	Stream responses, cache results, and move the local model to a separate inference service or drop it for Gemini only.
4	Security	Open access; API key kept in environment variables.	Add login (OAuth or JWT), input length limits, rate limiting and usage monitoring.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	User login, saved history and a progress dashboard with quiz score tracking	2 Months	Learners can continue where they stopped and see their improvement.
Phase 3	PDF and image input, multilingual support and voice input	4 Months	Makes EduGenie usable with real study material, in more languages, hands-free.
Phase 4	Gamification (badges, streaks), teacher dashboards, LMS integration (Moodle, Google Classroom) and a mobile app	6 Months	Boosts motivation, supports classrooms and offers learning anywhere.